Conditionals in JS

Lec-4

If Statement

- The **If Statement** in JS executes the code only when a condition is **true**.
- **Eg :**

```
let cartEmpty = true

if (cartEmpty){
   console.log('Your Cart is Empty');
}

// Output: Your Cart is Empty.
```

If you've to print single statement after passing it through a 'if' statement, alternatively it can be done as follows:

```
let cartEmpty = true
if (cartEmpty) console.log("Your Cart is Empty");

// Output: Your Cart is Empty.
```

```
let randomNum = 56
if (randomNum > 100){
   console.log("The number is greater");
}
// Nothing will get printed to console as the condition given in if statement her is turning out to be false.
```

Truthy Value and Falsy Value

Truthy	**true, "false", "0", 1 , [], {}**
Falsy	**false , 0, -0, "", null, NaN, undefined**

Else Statement

- The Else Statement is to specify a block of code to be executed **only when a if condition turns out to be false**.
- **Eg :**

```
let moneyPresent = false
if(moneyPresent){
   console.log("Account balance");
}
else{
   console.log("You don't have any balance in your account");
}

//Output: You don't have any balance in your account
```

Solve This Practice Question

Write a simple JavaScript program to check whether the person is **eligible to drive or not**.

Create 2 variables to store a person's **Age** and the **Country** which he's living in

If the person's **age is greater than or equal to 16** and **country is USA** , print ***"You're eligible to drive !!"***

else print ***"You're not eligible to drive !!"***

Ternary Operator

- Ternary Operator is a **shorthand** for writing **if-else conditional statements**.
- It is called **Ternary Operator** because it **takes three operands**.
- **Eg :**

Traditional if-else conditional

```
let health = 'Sick'
if(health.toLowerCase() === 'sick'){
   console.log('Go to doctor');
}
else{
   console.log("Enjoy at home");
}
```

Ternary Operator

```
console.log((health.toLowerCase() === 'sick') ? "Go to doctor" : "Enjoy at home");
```

Else if statement

- The Else if Statement is to specify a **new condition** if the **first condition is false**.
- **Eg** :

```
let foodItem = 'Hello'

if (foodItem.toLowerCase() === 'breakfast'){
    console.log("Idli Sambhar");
}
else if(foodItem.toLowerCase() === 'lunch'){
    console.log('Non veg Thali');
}
else if(foodItem.toLowerCase() === 'dinner'){
    console.log('Veg Thali');
}
else{
    console.log("Invalid input");
}
```

Solve This Practice Question

Imagine a home automation tool which automatically turns on **Heater, Fan** or **Air Conditioner** based on sensing the current temperature of the house.

The conditions to be followed are:

- If current temperature is below **18°C,** then **"Turn the Heater on"**
- If current temperature is between **18°C and 25°C (inclusive,** then **"Turn the Fan on"**
- If current temperature is **above 25°C,** then **"Turn the AC on"**

Switch Statement

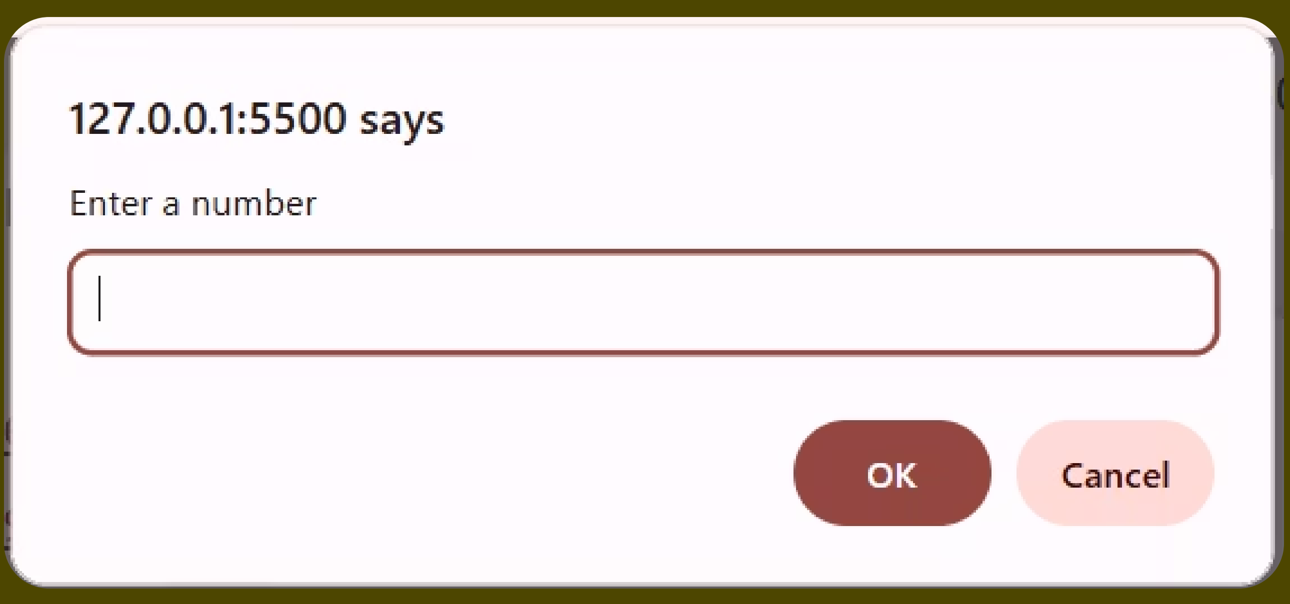
- Based on a condition, **switch** statement **selects one or more code blocks to be executed**.
- switch executes an code block **that matches an expression**.
- Switch acts as an **alternative to many if-else statements** hampering readability of the code.
- In switch statement we've to give an **break** keyword for **preventing a "fall-through"**.
- The **default** keyword specifies a block of code to run **if there is no case match**.
- **Without** the **break keyword**, the code will **keep executing the next case block including the default case block,** even when **values do not match** with the expression.

Note:

How to Accept user input in JS

- **prompt() :** This method displays a dialog box that prompts the user for input.
- This method **returns the input value** if the user clicks **"OK"**, otherwise it returns **null**
- **Eg** :

```
prompt("Enter a number")
```



- prompt() takes an input from user and **always store it in form of String**.

Methods to convert Strings into number

- **Number() :** Returns a number, converted from its argument.

```
let Stringnum = "100"
console.log(Stringnum); // 100
console.log(typeof Stringnum); // string
console.log(Number(Stringnum)); // 100
console.log(typeof Number(Stringnum)); // number
```

- **parseFloat() :** Parses a string and returns a floating point number.

```
let Stringnum = "69.99"
console.log(Stringnum); // 69.99
console.log(typeof Stringnum); // string
console.log(parseFloat(Stringnum)); // 69.99
console.log(typeof Number(Stringnum)); // number
```

- **parseInt() :** Parses a string and returns an integer.

```
let Stringnum = "69.99"
console.log(Stringnum); // 69.99
console.log(typeof Stringnum); // string
console.log(parseFloat(Stringnum)); // 69.99
console.log(typeof Number(Stringnum)); // number
```

- **Example for switch case** :

```
let Daynum = parseFloat(prompt("Enter a number"))
console.log(Daynum);

switch(Daynum){
  case 1:
    console.log("It's Sunday");
    break;
  case 2:
    console.log("It's Monday");
    break;
  case 3:
    console.log("It's Tuesday");
    break;
  case 4:
    console.log("It's Wednesday");
    break;
  case 5:
    console.log("It's Thursday");
    break;
  case 6:
    console.log("It's Friday");
    break;
  case 7:
    console.log("It's Saturday");
    break;
  default:
    console.log("Invalid input");
}
```

Solve This Practice Question

Imagine you're builiding a program for an tele caller of an bank application which does different tasks on basis of selection of different option by the user.

- Initially, print message to the user **"Welcome to the BANK OF INDIA"**
- Then print the message in following way:

  "Press 1 to know your account balance"

  "Press 2 to edit your account details"

  "Press 3 to verify your EKYC"

  "Press 4 to explore our loan options"

- Take input from user using **prompt()**
- Manipulate the input using switch case in following manner

If user inputs **1** print **"You've selected to view your account balance"**

If user inputs **2** print **"You've selected to edit your account details"**

If user inputs **3** print **"You've selected to verify your EKYC"**

If user inputs **4** print **"You've selected to explore our loan options"**

Solve these Questions on Conditionals

Question 1 — Form validator for a signup page

Imagine you're building the frontend of a user registration page. Before a new user is allowed to create their account, your JavaScript code needs to validate their details on the spot.

Create the following variables to represent the user's submitted data:

- A variable **username** storing a string value
- A variable **age** storing a number
- A boolean variable **agreedToTerms**

Now validate the data using conditionals in the following way:

- Using an **if statement**, check if **username** is **non-empty** — if it is **empty (falsy)**, print **"Username cannot be empty"**
- Using **if / else if / else**, check the **age**

— if age is **below 13**, **print "Too young to register"**

— if age is **between 13 and 17**, **print "Parent approval required"**

— if age is **18 or above**, **print "Welcome aboard!"**

- Use a **ternary operator** to print **"Terms accepted"** if **agreedToTerms is true**, otherwise **print "Must accept terms to continue"**

Test your code by changing the values of all three variables and observe how the output changes each time.

Question 2 — Delivery status tracker

Imagine you're building the **"Track My Order"** screen of an e-commerce app. The backend sends a numeric status code, and your job is to translate that code into a clear, human-readable message for the customer.

Use pro**mpt()** to ask the user: **"Enter your delivery status code (1–5):"**

Write a **switch statement** that maps each code to a message in the following way:

- If the user inputs **1**, **print "Order placed"**
- If the user inputs **2, print "Packed and dispatched"**
- If the user inputs **3, print "Out for delivery"**
- If the user inputs **4, print "Delivered successfully"**
- If the user inputs **5, print "Delivery failed — contact support"**
- For any **other input**, **print "Invalid status code — please check your order ID"**

Question 3 — Subscription plan upgrade prompt

Imagine you're building the **billing logic for a SaaS dashboard**. Based on the user's current subscription plan and their monthly usage, your program should decide whether to show them an upgrade prompt.

Use **prompt()** to ask: **"Enter your current plan (free / pro / enterprise):"**

Use **prompt() again** to ask: **"Enter your monthly usage in GB:"**

Write an if / **else if / else chain** to handle the following conditions:

- If the plan is **"free"** and usage is **above 2 GB**, print **"Upgrade to Pro for more storage"**
- If the plan is **"pro"** and usage is **above 50 GB,** print "**Consider upgrading to Enterprise"**
- If the plan is **"enterprise"**, print **"You're on our best plan!"**
- For any **other combination**, print **"Usage is within limits, no upgrade needed"**

Additionally, create a **boolean** variable **isBillingActive** and **set it yourself**. Use a ternary operator to print either **"Billing: Active"** or **"Billing: Inactive"** based on its value.

Test the full program by trying different combinations of plan, usage, and billing status.